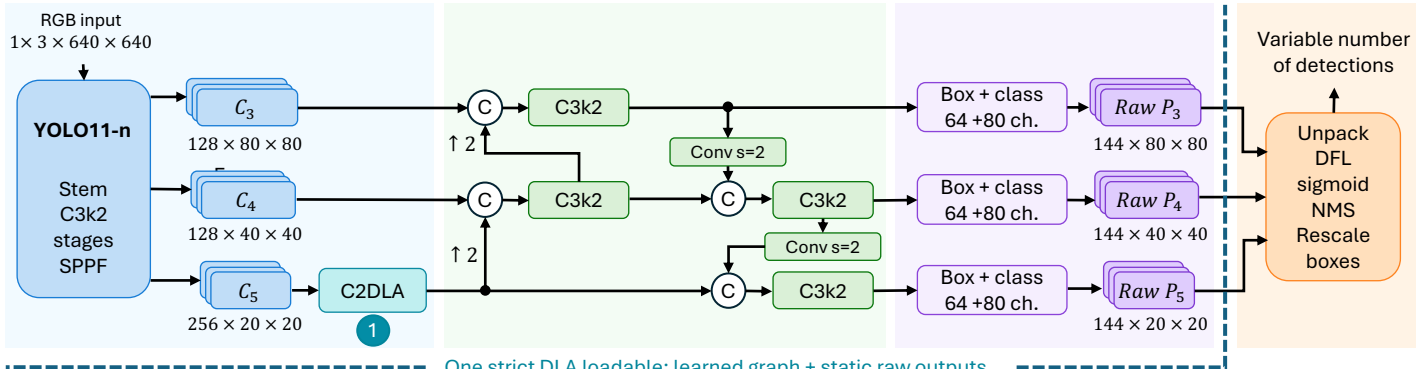
(a) Backbone
(b) Multi-scale neck
(c) DetectDLA
2
(d) CPU
RGB input
1× 3 × 640 × 640
YOLO11-n
Stem
C3k2
stages
SPPF
C_3
128 × 80 × 80
C_4
128 × 40 × 40
C_5
256 × 20 × 20
C2DLA
1
↑ 2
C
C3k2
Conv s=2
Box + class
64 +80 ch.
$Raw\ P_3$
144 × 80 × 80
$Raw\ P_4$
144 × 40 × 40
$Raw\ P_5$
144 × 20 × 20
Variable number
of detections
Unpack
DFL
sigmoid
NMS
Rescale
boxes
One strict DLA loadable: learned graph + static raw outputs
1 C2PSA → C2DLA
2 Raw export boundary
C channel concat
↑ 2 nearest upsampling
Dimensions omit batch B = 1